CYBERSRS

Secure Access Control and Network Monitoring for University Research Laboratories

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure access-control and network-monitoring system for university research laboratories containing sensitive equipment and research data. The system should allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions. Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours. The system should maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices. The solution should be secure, reliable, and manageable across multiple research buildings.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure access-control and network-monitoring system for university research laboratories containing sensitive equipment and research data.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The system should maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Status: confirmed

Evidence ids: E-004

Stakeholder source statement E-005

Detail: The solution should be secure, reliable, and manageable across multiple research buildings.

Status: confirmed

Evidence ids: E-005

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - provide secure access-control and network-monitoring capabilities for university research laboratories containing sensitive equipment and research data.

Statement: The system shall provide secure access-control and network-monitoring capabilities for university research laboratories containing sensitive equipment and research data.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A research laboratory with sensitive equipment and data

When: The system is deployed to monitor and control access

Then: Access to the lab systems is restricted to authorized users only, and network traffic is monitored for anomalies

Method: test

Observable evidence: Test capture of the resulting state

FR-002 - allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.

Statement: The system shall allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A user with defined roles: student, faculty, researcher, administrator

When: The user attempts to access a lab network system

Then: Access is granted only if the user's role and permissions are authorized by the system

Method: inspection

Observable evidence: Test capture of permission evaluation response showing role-based access decisions

FR-003 - Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours.

Statement: Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrative user with valid credentials and access rights

When: The administrator initiates a request to grant or revoke access to a system

Then: The administrator observes access permissions are updated

Method: demonstration

Observable evidence: Test capture of the resulting response

Given: An administrative user with valid credentials and access rights

When: The administrator reviews active sessions and monitors login activity

Then: The administrator sees a list of active sessions and login activity records

Method: analysis

Observable evidence: Test capture of the resulting state

Given: An administrative user with valid credentials and access rights

When: The system detects repeated failed authentication attempts or unauthorized access outside approved hours

Then: The system detects repeated failed authentication attempts or unauthorized access outside approved hours

Method: test

Observable evidence: Test capture of the observed result

FR-004 - The solution should be secure, reliable, and manageable across multiple research buildings.

Statement: The solution should be secure, reliable, and manageable across multiple research buildings.

Rationale: Implements stakeholder source E-005.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-005

Acceptance criteria

Given: A multi-building university environment with distributed lab networks

When: The system is configured to manage access and monitoring across all buildings

Then: Access policies are consistently enforced, and network monitoring remains functional without degradation in performance or reliability

Method: test

Observable evidence: Test capture of cross-building policy enforcement and monitoring continuity

3. Non-functional requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-004

4. Security, privacy and compliance

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

SEC-001 - maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Statement: The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: An administrator performs an access decision or action

When: The system records the event in an audit log

Then: The log entry includes timestamp, user identity, action type, target resource, and outcome

Method: test

Observable evidence: Test capture of audit log entries for access decisions and administrative actions

Given: A research device connects to a lab network via approved configuration

When: The system validates the device's identity and permissions

Then: The connection is allowed only if the device is university-managed or has an approved research device certificate

Method: demonstration

Observable evidence: Test capture of device onboarding and permission validation

Given: A student attempts to access a lab network from outside campus hours

When: The system enforces access restrictions based on time and location rules

Then: Access is denied with a message indicating the restriction, and an alert is generated for administrator review

Method: inspection

Observable evidence: Test capture of access denial events and corresponding alerts

Unresolved decisions: Thresholds for what constitutes unusual login activity; Definition of 'approved research device' and how it is validated

Related requirement ids: FR-001; FR-002; FR-003

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: SEC-001

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; SEC-001

7. Compliance and policy applicability matrix

Open decisions: Thresholds for what constitutes unusual login activity; Definition of 'approved research device' and how it is validated

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Access Control Engine

Detail: Proposed logical owner for FR-002, SEC-001, FR-004.

Status: proposed

Evidence ids: E-002; E-004; E-005

Requirement ids: FR-002; SEC-001; FR-004

Component name: Access Control Engine

Responsibility: The system shall allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices. The solution should be secure, reliable, and manageable across multiple research buildings.

Network Monitoring Module

Detail: Proposed logical owner for FR-001, SEC-001, FR-004.

Status: proposed

Evidence ids: E-001; E-004; E-005

Requirement ids: FR-001; SEC-001; FR-004

Component name: Network Monitoring Module

Responsibility: The system shall provide secure access-control and network-monitoring capabilities for university research laboratories containing sensitive equipment and research data. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices. The solution should be secure, reliable, and manageable across multiple research buildings.

Audit and Alert Service

Detail: Proposed logical owner for FR-003, SEC-001, FR-004.

Status: proposed

Evidence ids: E-003; E-004; E-005

Requirement ids: FR-003; SEC-001; FR-004

Component name: Audit and Alert Service

Responsibility: Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices. The solution should be secure, reliable, and manageable across multiple research buildings.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

From component: Access Control Engine

To component: Network Monitoring Module

Data or action: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

From component: Access Control Engine

To component: Audit and Alert Service

Data or action: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

From component: Network Monitoring Module

To component: Access Control Engine

Data or action: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate

research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

From component: Network Monitoring Module

To component: Audit and Alert Service

Data or action: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

From component: Audit and Alert Service

To component: Access Control Engine

Data or action: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

From component: Audit and Alert Service

To component: Network Monitoring Module

Data or action: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

Untrusted component: Access Control Engine

Trusted component: Network Monitoring Module

Enforcement: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

Untrusted component: Access Control Engine

Trusted component: Audit and Alert Service

Enforcement: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

Untrusted component: Network Monitoring Module

Trusted component: Access Control Engine

Enforcement: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

Untrusted component: Network Monitoring Module

Trusted component: Audit and Alert Service

Enforcement: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

Untrusted component: Audit and Alert Service

Trusted component: Access Control Engine

Enforcement: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004, SEC-001.

Status: proposed

Evidence ids: E-004; E-005

Requirement ids: FR-004; SEC-001

Untrusted component: Audit and Alert Service

Trusted component: Network Monitoring Module

Enforcement: The solution should be secure, reliable, and manageable across multiple research buildings. The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: students

Goal: The system shall allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.

Story: As students, I need the following behavior: The system shall allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.

Preconditions

1. A user with defined roles: student, faculty, researcher, administrator

Steps

1. The user attempts to access a lab network system

Success outcome: Access is granted only if the user's role and permissions are authorized by the system

Requirement ids: FR-002

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A research laboratory with sensitive equipment and data
2. When: The system is deployed to monitor and control access

Expected result: Expected: Access to the lab systems is restricted to authorized users only, and network traffic is monitored for anomalies

Evidence to record: Evidence: Test capture of the resulting state

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: A user with defined roles: student, faculty, researcher, administrator
2. When: The user attempts to access a lab network system

Expected result: Expected: Access is granted only if the user's role and permissions are authorized by the system

Evidence to record: Evidence: Test capture of permission evaluation response showing role-based access decisions

Test type: inspection

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: An administrative user with valid credentials and access rights
2. When: The administrator initiates a request to grant or revoke access to a system

Expected result: Expected: The administrator observes access permissions are updated

Evidence to record: Evidence: Test capture of the resulting response

Test type: positive

Id: TEST-004

Requirement ids: FR-003

Procedure

1. Given: An administrative user with valid credentials and access rights
2. When: The administrator reviews active sessions and monitors login activity

Expected result: Expected: The administrator sees a list of active sessions and login activity records

Evidence to record: Evidence: Test capture of the resulting state

Test type: analysis

Id: TEST-005

Requirement ids: FR-003

Procedure

1. Given: An administrative user with valid credentials and access rights
2. When: The system detects repeated failed authentication attempts or unauthorized access outside approved hours

Expected result: Expected: The system detects repeated failed authentication attempts or unauthorized access outside approved hours

Evidence to record: Evidence: Test capture of the observed result

Test type: negative

Id: TEST-006

Requirement ids: FR-004

Procedure

1. Given: A multi-building university environment with distributed lab networks
2. When: The system is configured to manage access and monitoring across all buildings

Expected result: Expected: Access policies are consistently enforced, and network monitoring remains functional without degradation in performance or reliability

Evidence to record: Evidence: Test capture of cross-building policy enforcement and monitoring continuity

Test type: positive

Id: TEST-007

Requirement ids: SEC-001

Procedure

1. Given: An administrator performs an access decision or action
2. When: The system records the event in an audit log

Expected result: Expected: The log entry includes timestamp, user identity, action type, target resource, and outcome

Evidence to record: Evidence: Test capture of audit log entries for access decisions and administrative actions

Test type: positive

Id: TEST-008

Requirement ids: SEC-001

Procedure

1. Given: A research device connects to a lab network via approved configuration
2. When: The system validates the device's identity and permissions

Expected result: Expected: The connection is allowed only if the device is university-managed or has an approved research device certificate

Evidence to record: Evidence: Test capture of device onboarding and permission validation

Test type: positive

Id: TEST-009

Requirement ids: SEC-001

Procedure

1. Given: A student attempts to access a lab network from outside campus hours
2. When: The system enforces access restrictions based on time and location rules

Expected result: Expected: Access is denied with a message indicating the restriction, and an alert is generated for administrator review

Evidence to record: Evidence: Test capture of access denial events and corresponding alerts

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Access control

Detail: Rules and enforcement governing which principals may perform actions on resources.

Status: proposed

Evidence ids: E-001

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-004

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-004

Device isolation

Detail: Restriction of a device's network access; the source determines whether and by whom this action is permitted.

Status: proposed

Evidence ids: E-004

14. SDLC recommendation and workflow

Selected model: Agile

Justification: The selection of Agile as the required model is based on a uniform score of 40 across all candidates, with no differentiating input evidence provided to favor one methodology over another. Given the absence of explicit factors such as budget, delivery timelines, failure consequences, feedback mechanisms, legacy systems, regulation, schedule constraints, stability needs, uncertainty levels, or verification requirements, no candidate model can be objectively ranked for suitability. The provisional nature of this decision reflects an incomplete evidence base. All models receive equal score meaning—suitability points out of 100—not probabilities. Without evidence to differentiate between Agile and other methodologies (e.g., DevSecOps, Spiral, V-Model, Waterfall, Hybrid, Incremental), the choice remains arbitrary and unsupported by input data. The requirement for secure access control, role-based permissions, real-time monitoring, audit logging, and cross-building manageability aligns with iterative development principles but does not inherently mandate Agile over other models. Unknown factors such as regulation or legacy systems could influence model selection if addressed; however, these are currently unaccounted for in the input evidence. Therefore, this decision is grounded only in equal suitability scores and lacks evidentiary differentiation.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall provide secure access-control and network-monitoring capabilities for university research laboratories containing sensitive equipment and research data.; FR-002: The system shall allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.; FR-003: Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours.; FR-004: The solution should be secure, reliable, and manageable across multiple research buildings.; SEC-001: The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.
2. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against SEC-001: The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; SEC-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall provide secure access-control and network-monitoring capabilities for university research laboratories containing sensitive equipment and research data.; FR-002: The system shall allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.; FR-003: Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours.; FR-004: The solution should be secure, reliable, and manageable across multiple research buildings.; SEC-001: The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.
3. In an authorized isolated test environment, execute TEST-001: Given: A research laboratory with sensitive equipment and data When: The system is deployed to monitor and control access Expected: Access to the lab systems is restricted to authorized users only, and network traffic is monitored for anomalies
4. In an authorized isolated test environment, execute TEST-002: Given: A user with defined roles: student, faculty, researcher, administrator When: The user attempts to access a lab network system Expected: Access is granted only if the user's role and permissions are authorized by the system
5. In an authorized isolated test environment, execute TEST-003: Given: An administrative user with valid credentials and access rights When: The administrator initiates a request to grant or revoke access to a system Expected: The administrator observes access permissions are updated
6. In an authorized isolated test environment, execute TEST-004: Given: An administrative user with valid credentials and access rights When: The administrator reviews active sessions and monitors login activity Expected: The administrator sees a list of active sessions and login activity records
7. In an authorized isolated test environment, execute TEST-005: Given: An administrative user with valid credentials and access rights When: The system detects repeated failed authentication attempts or unauthorized access outside approved hours Expected: The system detects repeated failed authentication attempts or unauthorized access outside approved hours

8. In an authorized isolated test environment, execute TEST-006: Given: A multi-building university environment with distributed lab networks When: The system is configured to manage access and monitoring across all buildings Expected: Access policies are consistently enforced, and network monitoring remains functional without degradation in performance or reliability
9. In an authorized isolated test environment, execute TEST-007: Given: An administrator performs an access decision or action When: The system records the event in an audit log Expected: The log entry includes timestamp, user identity, action type, target resource, and outcome
10. In an authorized isolated test environment, execute TEST-008: Given: A research device connects to a lab network via approved configuration When: The system validates the device's identity and permissions Expected: The connection is allowed only if the device is university-managed or has an approved research device certificate
11. In an authorized isolated test environment, execute TEST-009: Given: A student attempts to access a lab network from outside campus hours When: The system enforces access restrictions based on time and location rules Expected: Access is denied with a message indicating the restriction, and an alert is generated for administrator review
12. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
13. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against SEC-001: The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; SEC-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against SEC-001: The system shall maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; SEC-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure access-control and network-monitoring system for university research laboratories containing sensitive equipment and research data.

Id: E-002

Source: description

Quote: The system should allow only authorized students, faculty, researchers, and lab administrators to access designated lab networks and systems based on role and approved permissions.

Id: E-003

Source: description

Quote: Administrative users should be able to grant or revoke access, review active sessions, monitor unusual login activity, and receive alerts for repeated failed authentication, unauthorized access attempts, and access outside approved lab hours.

Id: E-004

Source: description

Quote: The system should maintain audit logs of access decisions and administrator actions, isolate research lab networks from general student and guest networks, and support both university-managed and approved research devices.

Id: E-005

Source: description

Quote: The solution should be secure, reliable, and manageable across multiple research buildings.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7412

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7242

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:2 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 2

Page or section: text chunk 3

Relevance score: 0.7057

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

kb-c1e0239af9dd46bd:2 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 2

Page or section: text chunk 3

Relevance score: 0.6897

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.7083

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Security, privacy and compliance: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	UC-001, TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	TEST-003, TEST-004, TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-004	E-005	TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-001	E-004	TEST-007, TEST-008, TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28